



Cohorts, Copper, Crisis

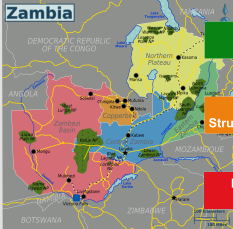
Zambia's Demographic Trajectory across 3 Regimes (1964-2024)

Image: Reuters/
S. Henry
via DW

Image:
Wikitravel

Image: pexels

Motivation and World System's Theory⁹



Regime 1 (1964-1991):
State Capitalism

Regime 2 (1991-2000):
Structural Adjustment Programs

Regime 3 (2000-Present):
Foreign Capital

- Colonial independence + Nationalization (semi-peripheral status aspiration)
- Efforts to retain surplus copper value, invest revenues domestically

- World Bank/IMF impose austerity measures & privatization through SAPs
- Re-peripheralization: foreign firms buy up largest copper mines in fire sale⁷
- Neoliberal reforms weakened state developmental functions⁸

- Multinational ownership of copper extraction (first Western⁶, later Chinese⁵) → bulk of copper revenues leave country
- 2021³: Gov seeks IMF loan to acquire and re-nationalize the Mopani mine

Research Question: Is Zambia's Demographic Transition associated with copper industry privatization?

World System's and Dependency Theory - extraction of surplus value from the factors of production (i.e., raw resources, land, labor) destabilizes and/or prevents direct domestic investment in the economies of peripheral countries

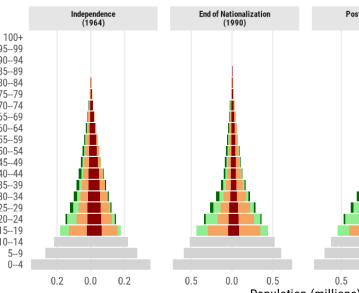
Zambia provides a theoretical test case: world's 7th largest copper reserve, serving 69% of its total exports in 2022¹
Recent scenarios project 1.4-2m tonnes of output by 2030, positioned to be a top-5 global copper producer³

Paradoxical finding: education expanded dramatically, so demographic transition expected to follow

Yet fertility remains high despite improvement in mortality, as surmised by increased Life Expectancy

Zambia's Demographic & Educational Evolution over Time

Population pyramids by age, sex, and educational attainment (1965-2020)



Data: Wittgenstein Centre wdc (K.C et al, 2024) | Color stratified by highest educational attainment
Image: nsc.org

Year	% No Edu	% Secondary+
1965	34.9	8.1
1990	22.2	8
2000	18.1	7.9
2020	11.7	9.3

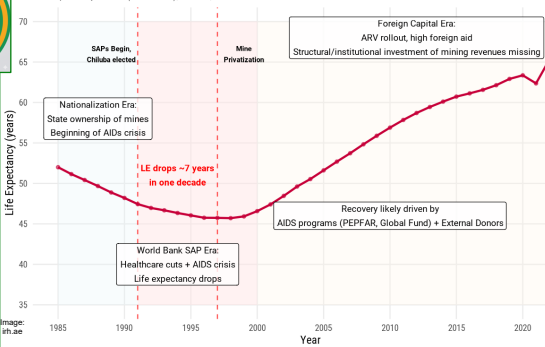
Population soared by **416%** (3.7m to 19.1m),

yet pyramid base (ages under 15) remained wide in 2020

Observation suggests fertility transition is stalling

Zambia's Life Expectancy: Before, During, and After Structural Adjustment

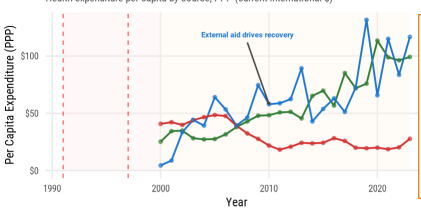
Life expectancy at birth (both sexes), Zambia, 1985-2023



Data: UN World Population Prospects (2024) | Shaded regions indicate political-economic regimes

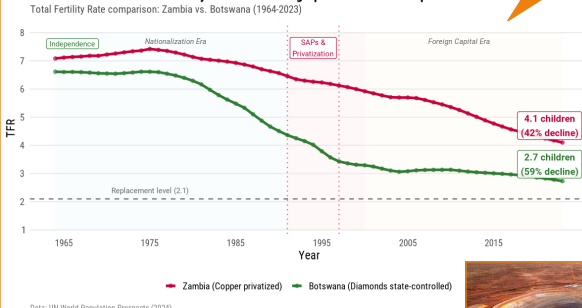
Healthcare Financing: High Donor Dependency

Health expenditure per capita by source, PPP (current international \$)



Resource Wealth and the Fertility Decline: A Demographic Transition Comparison

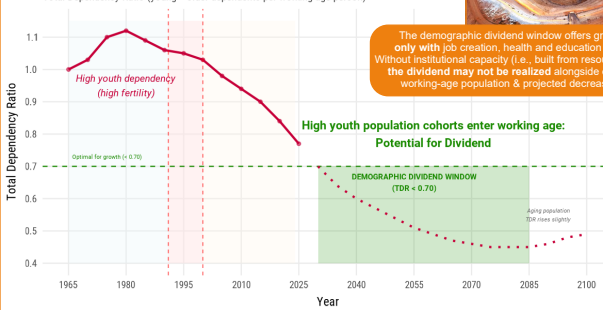
Total Fertility Rate comparison: Zambia vs. Botswana (1964-2023)



Data: UN World Population Prospects (2024)

Zambia's Demographic Dividend Window: 2030-2085

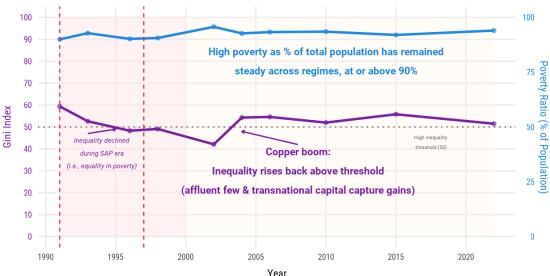
Total Dependency Ratio (young + older dependents per working-age person)



Data: Wittgenstein Centre wdc (K.C et al, 2024) | Solid line = historical, dotted = projected

Inequality and Poverty Trends, 1991-2022

Gini Index (left axis) and Poverty Headcount Ratio (right axis)



Data: World Bank WDI (2025) | Gini Index on 0-100 scale, higher = more inequality | Poverty threshold calculated at \$8.30 a day (2021 PPP)

Privitization⁷

(1997-2000)

Copper mines sold for:

- Konkola: \$25m

- Kansanshi: \$28

- Luanshya: \$35m

Total: ~\$125m assets,

By 2006, ~\$3b revenues

generated annually for

foreign companies,

while Zambian gov

received \$70m (2.3%)

In 2021, Zambian government announced to re-nationalize Mopani through more borrowing - after 20 years of profit extraction

Demographic Transition theory predicts reduced fertility + longer life expectancy = economic growth,

but education enhances adaptive capacity *only if* complemented by health infrastructure and economic opportunity

Because TFR remains high, **first demographic transition stalled despite gains in education**, revenues flowed to foreign capital

Scan me for citations



Image: mining-technology.com

